Solution

Problems: Time-Domain System Descriptions Review

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1. Which of the following systems with input $x(t)$ and output $y(t)$ are stable? Select all that apply.

a)

$$\frac{d}{dt}y(t) - 0.5y(t) = 2x(t) + \frac{d}{dt}x(t)$$

b)

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau, \quad h(\tau) = \cos(\pi\tau)u(\tau+1)$$

c)

$$y(t) = \log(x(t+1))$$

d)

$$\frac{d}{dt}y(t) + 3y(t) = \frac{d}{dt}x(t)$$

e) none are stable

a) Differential eqn. For stability all roots of the characteristic eqn must have negative real parts.

$r - 0.5 = 0 \Rightarrow r = 0.5$ not stable

b) $y(t) = x(t) * h(t)$ so we can think of $h(t)$ as the impulse response of an LTI system. For stability $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$

Here $\int_{-\infty}^{\infty} |h(t)|\,dt = \int_{-1}^{\infty} |\cos(\pi t)|\,dt$ which is not bounded. Unstable.

c) We require $|y(t)| < \infty$ for all $|x(t)| < M_x$ (BIBO)

Clearly $x(t+1) = 0$ is bounded, but leads to unbounded output since $\log(0) \to -\infty$ Unstable

d) See answer to a). Here char eqn $r + 3 = 0$, so $r = -3$. Root has negative real part.

2. Which of the following systems with input $x(t)$ and output $y(t)$ are causal? Select all that apply.

a)

$$\frac{d}{dt}y(t) - 0.5y(t) = 2x(t) + \frac{d}{dt}x(t)$$

b)

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau, \quad h(\tau) = \cos(\pi\tau)u(\tau+1)$$

c)

$$y(t) = \log(x(t+1))$$

d)

$$\frac{d}{dt}y(t) + 3y(t) = \frac{d}{dt}x(t)$$

e) none are causal

a) Differential equations represent causal systems as we have solved them – assuming initial conditions at $t=0^-$ and finding the output forward from $t=0$

b) $y(t) = x(t) * h(t)$, $h(t)$ is impulse response of LTI system. For causality we require $h(t) = 0$ for $t<0$ Here $h(t) \neq 0$ for $-1<t<0$, so not causal.

c) output at time t depends on input at a future time, $t+1$. not causal.

d) See answer to a)

3. Which of the following systems with input $x(t)$ and output $y(t)$ are time-invariant? Select all that apply.

a)

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau, \quad h(\tau) = e^{-\tau}\sin(\pi\tau)u(\tau-1)$$

b)

$$\frac{d}{dt}y(t) - 0.5y(t) = 2tx(t) + \frac{d}{dt}x(t)$$

c)

$$y(t) = (t+1)^2\cos(x(t+1))$$

d)

$$\frac{d}{dt}y(t) + 3y(t) = \frac{d}{dt}x(t)$$

e) none are time invariant

a) $y(t) = x(t) * h(t)$ is a linear, time-invariant system for any t.

b) On RHS we see the time-varying coefficient $2t$ multiplying $x(t)$. Time varying system.

c) Compare $y_1(t) = f(x(t-a))$ to $y_2(t) = f(x(t))\Big|_{t=t-a}$

Time invariance requires $y_1(t) = y_2(t)$ for all shifts a.

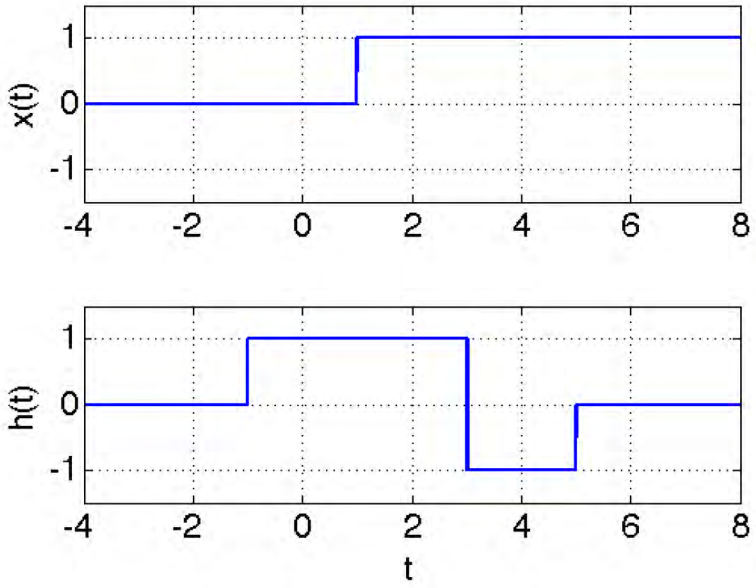
$y_1(t) = (t+1)^2\cos(x(t-a+1))$ $\quad y_2(t) = (t-a+1)^2\cos(x(t-a+1))$

Clearly $y_1(t) \neq y_2(t)$ for all a due to $(t+1)^2$ term

Time varying.

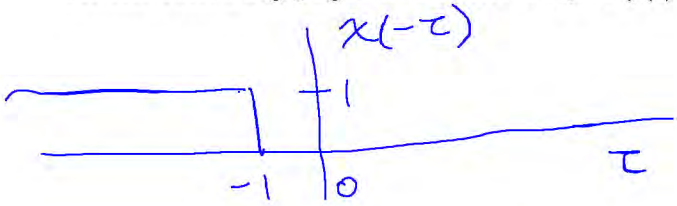
d) This is a constant coefficient differential equation $\Rightarrow$ time-invariant system

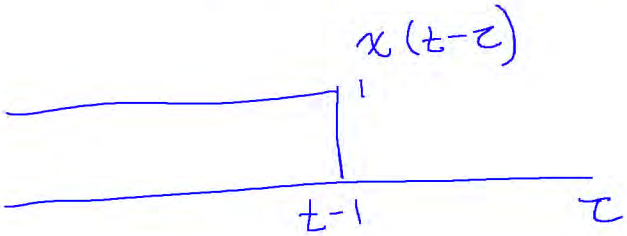
4. Consider a system with impulse response $h(t) = u(t+1) - 2u(t-3) + u(t-5)$ and input signal $x(t) = u(t-1)$, as shown below. Note the the signals continue outside the time

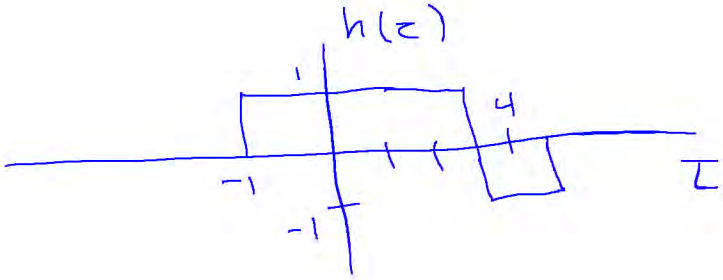


$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)\,d\tau$$

ranges shown by holding the values at the boundary of the graphs. Answer the questions on the following pages about the output $y(t)$ of this system.





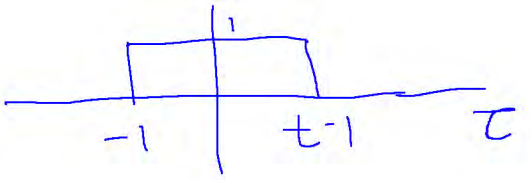


a) $y(t)$ is constant for $t <$ 0 and on this range $y(t) =$ 0.

$x(t-\tau)h(\tau) = 0$ for $t < 0$

b) As t increases beyond the answer to the previous question, $y(t)$ is (increasing, decreasing, constant) until $t =$ _____.

$x(t-\tau)h(\tau)$

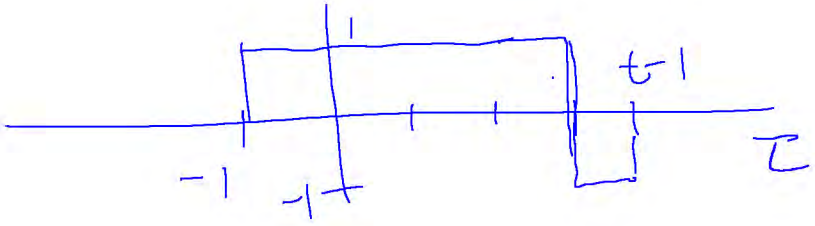


area increases as $t \uparrow$

t valid for $0 < t < 4$

c) As t increases beyond the answer to the previous question, $y(t)$ is (increasing, decreasing, constant) until $t =$ 6.

$x(t-\tau)h(\tau)$



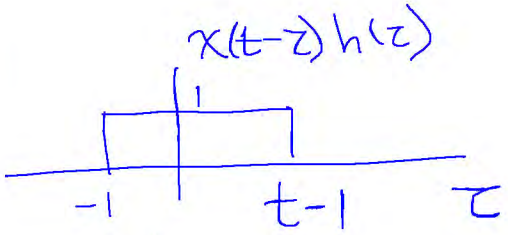
area decreases as $t \uparrow$

t valid for $4 < t < 6$

d) Find $y(t)$.

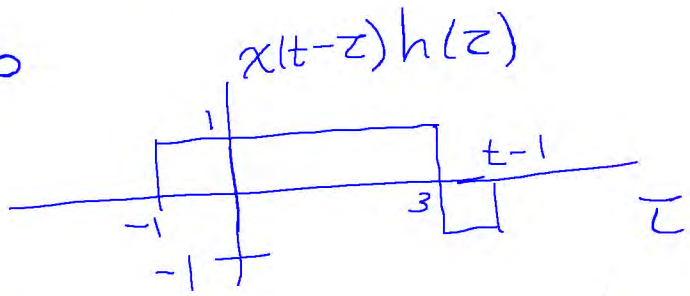
for $t<0$ $x(t-\tau)h(\tau)=0$ so $y(t)=0$

$0<t<4$



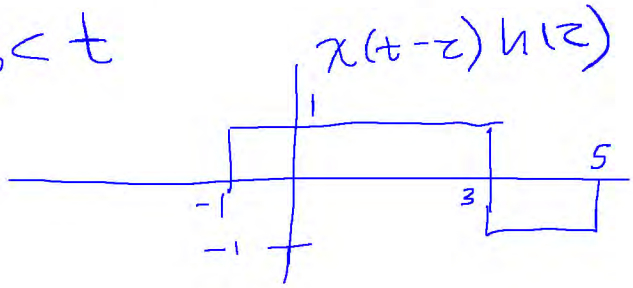
$$y(t)=\int_{-1}^{t-1} 1\,d\tau = t$$

$4<t<6$



$$y(t)=\int_{-1}^{3} 1\,d\tau + \int_{3}^{t-1}(-1)\,d\tau = 4-(t-4) = 8-t$$

$6<t$



$$y(t)=\int_{-1}^{3} 1\,d\tau - \int_{3}^{5} 1\,d\tau = 4-2=2$$

$$y(t)=\begin{cases} 0, & t<0 \\ t & 0<t<4 \\ 8-t & 4<t<6 \\ 2 & 6<t \end{cases}$$

5. A system with input $x[n]$ and output $y[n]$ is described by the difference equation

$$y[n] - \frac{1}{4}y[n-1] + ay[n-2] = x[n] + bx[n-1]$$

If the input is $x[n] = u[n]$ the output of the system is found to be $y[n] = 1 + \left(\frac{1}{2}\right)^n + 2\left(-\frac{1}{4}\right)^n$ for $n \geq 0$.

$y_s[n]$ $y_t[n]$

a) Enter the value $a =$ $-1/8$.

$y_t[n] = c_1(r_1)^n + c_2(r_2)^n$ where r_1, r_2 are roots of the characteristic equation. From $y[n]$ we know $r_1 = 1/2$ $r_2 = -1/4$ So $(r - 1/2)(r + 1/4) = 0$ is char eqn, or $r^2 - 1/4 r - 1/8 = 0$. Diff eqn implies char eqn is $r^2 - 1/4 r + a = 0$. Therefore $a = -1/8$

b) Enter the value $b =$ $-3/8$. $x[n] = z^n$ for $z = 1$

$\Rightarrow y_s[n] = H(1)\, 1^n \Rightarrow H(1) = 1$

from Diff eqn $H(z) = \dfrac{1 + b\, z^{-1}}{1 - 1/4\, z^{-1} - 1/8\, z^{-2}}$

$H(1) = \dfrac{1+b}{5/8} = 1 \Rightarrow b = \dfrac{5}{8} - 1 = -3/8$

c) Find the initial conditions $y[-1] =$ -5 and $y[-2] =$ -30.

Note $y[n]$ only applies for $n \geq 0$ $\quad y[0] = 1+1+2 = 4$

$y[1] = 1 + \frac{1}{2} - 2(\frac{1}{4}) = 1$

Use the difference equation to go backwards

$$y[n-2] = (x[n] - \tfrac{3}{8}x[n-1] - y[n] + \tfrac{1}{4}y[n-1])(-8)$$

$$= -8x[n] + 3x[n-1] + 8y[n] - 2y[n-1]$$

$n=1$: $\quad y[-1] = -8x[1] + 3x[0] + 8y[1] - 2y[0]$

$$= -8 + 3 + 8 - 2(4) = -5$$

$n=0$ $\quad y[-2] = -8x[0] + 3x[-1] - 8y[0] - 2y[-1]$

$$= -8 + 3\cdot 0 - 8\cdot 4 - 2(-5) = -30$$

d) Suppose the input is now $x[n] = 10(1/8)^n u[n]$. The steady-state component of the output is $y_s[n] = Ap^n$ where $A =$ $20/9$ and $p =$ $1/8$.

$x[n] = \alpha z^n \longrightarrow y_s[n] = H(z)\alpha z^n$

$z = \frac{1}{8}$, $\alpha = 10$

$$H(z) = \frac{1 - \frac{3}{8}z^{-1}}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}}$$

$$H(\tfrac{1}{8}) = \frac{1 - \frac{3}{8}\cdot 8}{1 - \frac{8}{4} - \frac{64}{8}} = \frac{-2}{1-2-8} = \frac{2}{9}$$

$$y_s[n] = \underbrace{\frac{20}{9}}_{A}\left(\underbrace{\tfrac{1}{8}}_{p}\right)^n$$